Regulatory Variant Report

AlphaGenome + DeepSeek Agent Analysis 2026-06-03 · regvar v0.1.0

chr17:48728343:C>T

chr17$_r$s138213197Ĺ MIR3185 **Unranked**

Scoring Summary

Assay	Gene	Biosample	Raw	Quantile
RNA-seq	MIR3185	myometrial cell	-0.05	**-1.00**
RNA-seq	MIR3185	GM23248	-0.05	**-1.00**
RNA-seq	MIR3185	M059J	-0.06	**-1.00**
RNA-seq	HOXB13	mesangial cell	-0.05	**-1.00**
RNA-seq	HOXB13	Caki2	-0.06	**-1.00**
RNA-seq	MIR3185	A172	-0.06	**-1.00**
RNA-seq	MIR3185	NCI-H460	-0.05	**-1.00**
RNA-seq	MIR3185	H4	-0.05	**-1.00**
RNA-seq	MIR3185	RPMI7951	-0.05	**-1.00**
RNA-seq	MIR3185	HFFc6	-0.05	**-1.00**
RNA-seq	HOXB13	bronchus fibroblast of lung	-0.05	**-1.00**
RNA-seq	MIR3185	SK-MEL-5	-0.06	**-1.00**
RNA-seq	MIR3185	AG04450	-0.05	**-1.00**
RNA-seq	MIR3185	SK-MEL-5	-0.05	**-1.00**
RNA-seq	MIR3185	mesenchymal stem cell	-0.05	**-1.00**

Interpretation

chr17:48728343:C>T (rs138213197) — OR = 3.85 | HOXB13** The strongest combined signal. This variant maps near **HOXB13**, a cardinal prostate lineage transcription factor. The alternate allele (T) is predicted to *reduce* HOXB13 and MIR3185 expression across diverse cell types (mesangial cells, Caki2 renal carcinoma, bronchus fibroblasts, mesenchymal stem cells — quantile scores up to 0.99980). The OR of 3.85 is the highest in the panel by a large margin. >

Predicted mechanism: The variant disrupts a regulatory element controlling HOXB13, reducing its expression in stromal and epithelial compartments. Given that HOXB13 is a pioneer factor in prostate identity and is recurrently mutated in prostate cancer (G84E), expression-level perturbation in the tumour microenvironment could alter AR-driven transcription and stromal-epithelial crosstalk.